1st lecture

Monday, August 24, 2026 9:49 AM

Math 6660 - FEMs

Context:

5610: interp, int, ODEs

5620: Numerical Lin. Algeb

6600 - FEMs

6670 → FVMs

5170 → FDMs



why FEM?

Original problem is continuous

(1) Theoritical:

FDM: approximite the solution to a discrete number of points;

FEM: keep orginal continous problem (PDE), but replace (approx) function

spaces.

All theory of PDE carried to FEM

② practical reasons:

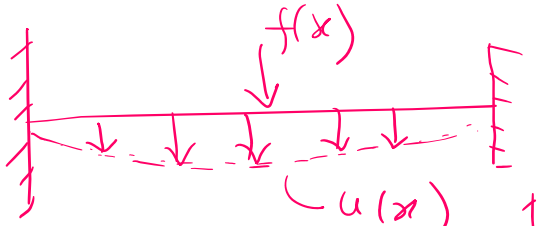
→ can be formulated ~~very~~ on unstructured grid

→ lots of flexibility in choice of function spaces

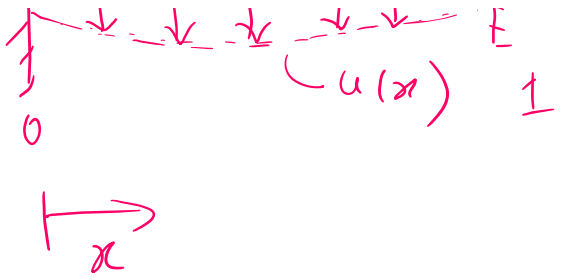
→

③ Note: much of theory of FEMs, is based on elliptic PDE theory. Parabolic PDE are ok. Hyperbolic PDEs are more challenging

Two point BVP? (1D version of poission)



$\rightarrow -\left(a(x)\,u'\right)' \equiv f(x)$



$\rightarrow -(a(x)u')' = f(x)$

$0 < x < 1$

$\rightarrow u(0) = u(1) = 1$

$u :=$ "displacment"
$f :=$ "external load" } given
$a =$ "modulus of elasticity" } given

Assume: $0 < a_1 \leq a(x) \leq a_2 < \infty$

$\forall \rightarrow x \in [0,1]$

Note $\mathcal{L}(u) = -(au')'$ with $u(0) = u(1) = 0$

(self adjoint operator) ~~is~~ a linear differential operator ($\mathcal{L}(u)$ is self-adjoint

$$\int_0^1 v\mathcal{L}(u)\,dx = \int_0^1 v(au')'\,dx$$

$v' \rightarrow u'$ $\quad a = 1$

$$= \int_0^1 v' a u' dx - \left[v a u' \right]_{x=0}^{x=1}$$

assume $u(0) = u(1) = 0$ and $v(0) = v(1) = 0$

$$= \int_0^1 v' a u' dx$$

$$= -\int_0^1 u(a'v') dx + \left[u a v' \right]_{x=0}^{x=1}$$

$$= -\int_0^1 u (av)' dx = \int_0^1 u \, \mathcal{J}(v) dx$$

$$\therefore \int_0^1 v \, \mathcal{J}(u) dx = \int_0^1 u \, \mathcal{J}(v) dx$$

(v: test function)

This is ~~not~~ what it means by to be self-adjoint

Note: self adjoint differential operator $\mathcal{J}$

Note: self adjoint differential operator $\mathcal{L}$ is analogous to symmetric matrices

Note: $\mathcal{L}(w) = 0$, $w(0) = w(1) = 0$

$\Leftrightarrow$ only solution that satisfy $\mathcal{L}$ & B.C.

$w = 0$

$\therefore$ if $\mathcal{L}$ is "invertible" $\Rightarrow$ for a given a & f there exist a unique u and f.

Note: if $0 < a < \alpha$ and f are sufficiently smooth

need: $a \in C^1(0,1)$ and $f \in C[0,1)$

then $u \in C^2(0,1)$

then, $u \in C^2(0,1)$

Variational Formulation:

Variational Formulation:

Multiply our BVP by a test function and integrate over the domain $[0, 1]$

$$-\int (au')' v \, dx = \int_0^1 f v \, dx$$

(f: force)

Then integrate by parts

$$\int_0^1 au'v' \, dx - [au'v]_0^1 = \int_0^1 f v \, dx$$

assume, v satisfy BCs:

$$v(0) = v(1) = 0$$

In particular, define solution space,

$$V = \left\{ v : \int_0^1 v^2 dx < \infty \; ; \int_0^1 [v']^2 dx < \infty, \right.$$

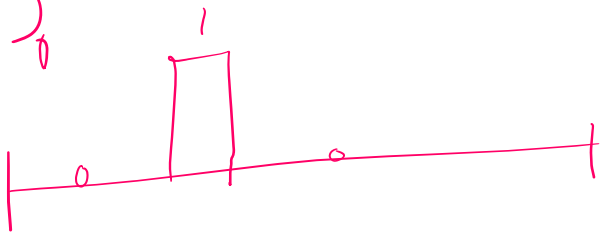
$$\left. v(0) = v(1) = 0 \right\}$$

Note:

f ... v are continous

functions in V are continous

(1) $\int_0^1 v^2 dx < \infty$ admits discontinous :



(2) $\int_0^1 [v']^2 dx$ does not allow

discontinous in V.

2nd lecture

2 point BVP

$-(au')' = f$ on fixed domain $0 < x < 1$

$u(0) = u(1) = 0$

find $u \in C^2(0,1)$

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variational formulation

$V = \{ v : \int_0^1 v^2 dx < 0 \longrightarrow$ defining a space

$v(0) = v(1) = 0 \}$

$$\int_0^1 (v')^2 dx < \alpha, \quad v(0) = v(1) = 0 \Big\}$$

Note: functions on V are continuous ($v \in C^0(0,1)$)

ex of discontinuous

$$H(x) = \begin{cases} 1 & \frac{1}{4} < x < \frac{3}{4} \\ 0 & \text{otherwise} \end{cases}$$

↳ not continuous

$H \notin C^0(0,1)$

but

$H \in C^{-1}(0,1)$

↳ finite jump

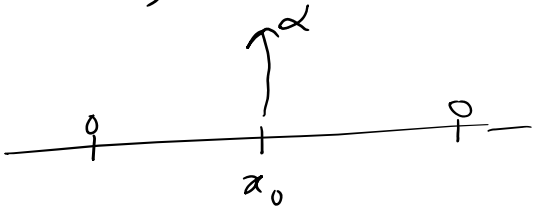
So,

$$\int_0^1 H^2 dx = \int_{1/4}^{3/4} dx = \frac{1}{2}$$

Now, the derivative,

$$H'(x) = \delta\left(x - \frac{1}{4}\right) - \delta\left(x - \frac{3}{4}\right)$$

$\delta(x - x_0)$:



δ property

$$\int_{x_0-\epsilon_1}^{x_0+\epsilon_2} \delta(x-x_0)\,dx = 1$$

for any $\epsilon_1, \epsilon_2 > 0$

Another one

$$\int_{x_0-\epsilon_1}^{x_0+\epsilon_2} f(x)\,\delta(x-x_0)\,dx = f(x_0)$$

f ~ continuous function

Now,

$$\int_0^1 (H')^2\,dx = \int_0^1 \left(\delta_{1/4} - \delta_{3/4}\right)^2 dx$$

$$= \int_0^1 \left(\delta_{1/4}^2 - 2\delta_{1/4}\delta_{3/4} + \delta_{3/4}^2\right) dx$$

Here, $\Rightarrow \int_0^1 \delta\left(x-\frac{1}{4}\right)\delta\left(x-\frac{1}{4}\right)dx = +\infty$

$\therefore H(x) \notin V$ because H' is not square

integrateable

Variational problem:

Find $u \in V$ such that

$$\int_0^1 a u' v' \, dx = \int_0^1 f v \, dx$$

for all $v \in \hat{V}$.

Note: u is the trial function

v are the test function

In this case u & v comes from the same function space, $\therefore V$

Q: How are (2 pt BVP) and (Vari problem) related?

→ BVP problem require $u \in C^2(0,1)$

→ Variational problem only need $u \in V$

→ Variational problem only need $u \in V$

claim 1:

If variational problem has a solution it is unique.

proof: (by contradiction)

assume ∃ 2 different solu: $u \neq U$

$$\int_0^1 a u' v' \, dx = \int_0^1 f v \, dx \quad \Big| \quad \int_0^1 a U' v' dx = \int_0^1 f v dx$$

Then, subtract:

$$\Rightarrow \int_0^1 a (u-U)' v' dx = 0 \quad \text{take, } v = u - U \in V$$

Now

$$\int_0^1 a \underbrace{\left((u-U)' \right)}_{\downarrow} dx = 0$$

$$(u-U)' = 0 \quad \text{on } (0,1)$$

$$\downarrow\downarrow$$

V

$u = U + c$

but, by BCs, $\therefore c = 0 \Rightarrow u = U$ (proved)

Claim 2 Solu to the BVP is a solution to variational problem

Proof: BVP has unique solution ~~as~~ is $u \in C^2(0,1)$

and $u(0) = u(1) = 0$

If $u \in C^2(0,1)$ + BCs $\Rightarrow u \in V$

Then BVP & variational problem are equivalent via int-by-parts

Claim 3: If $f \in C^0(0,1)$, and $a \in C^1(0,1)$, and

Lemma: If

$a \in C^1(0,1)$, and

$0 < a_1 \le a(x) \le a_2 < \infty$

$\forall \quad x \in (0,1)$

Then solution to variational problem is a solution ~~to~~ to BVP.

proof:

assume for the moment that $u \in C^2(0,1)$

$$\int_0^1 \left(a u' v' - f v \right) dx = 0$$

reverse int-by-parts:

$$\int_0^1 \underbrace{\left[(au')' + f \right]}_{S(x)} v \, dx = 0$$

Goal: to show $S(x) = 0$ for all $x \in (0,1)$

(a) Assume $S(\xi) = [(au')' + fu]\Big|_{x=\xi} > 0$

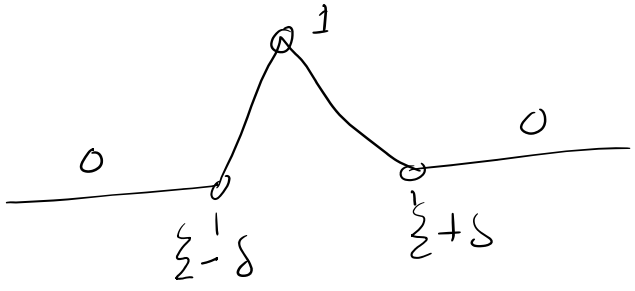
(proof by contradiction)

at some point $\xi \in (0,1)$

Note: that $S(x)$ is continuous in x.

By continuity: $S(x) > 0$ for $x \in [\xi - \delta, \xi + \delta]$ for some $\delta > 0$

choose v to be the following "hat" function



but $\int_0^1 S(x)\, v(x)\, dx > 0$

which is a contradiction.

$\therefore$ $S(x)$ cannot be positive in $(0,1)$

b) Redo this argument w/ $S(\xi) < 0$

b) Recall this ≥ 0

$\Rightarrow$ $S(x)$ cannot be negative in $(0,1)$

When can we expect $u \in C^2(0,1)$?

(if $f \in C^0(0,1)$, $a \in C^1(0,1)$ and

$*$ $0 < a_1 \leq a(x) \leq a_2 < \infty$ $\forall x \in (0,1)$)

Note:

(1) If $*$ is satisfied: $\rightarrow$ u is a strong solution of 2-pt BVP

(2-pt BVP) $\equiv$ (variational problem)

(2) If $*$ is not satisfied,

2-pt BVP will not have a solu.

but variational problem may still

have a solution. $\rightarrow$ u (if exist) is

a weak solu of

(2-pt BVP)